Scientific Analysis of Moral Judgments using Tobit and Cluster-Aware Non-Parametric Robustness Models

Automated Research Pipeline

March 30, 2026

1 Dataset and Sample Description

The empirical foundation of this project rests on two primary experimental datasets: FLORIDA and BUC. The sample consists of students from the Bucaramanga Campus. These datasets capture incentivized moral judgments from distinct socio-economic contexts. Participants were presented with standardized negotiation scenarios where a negotiator's decision resulted in varying degrees of payoff for themselves, their own group, and a victim group. Under Option 2: judgment-level relational modeling , each judgment is treated as relational and linked both to a descriptive victim x negotiator scenario shorthand and to hypothesis-specific predictors for judged negotiator status, decision outcome, and additional relational controls.

2 Datacard and Variable Definitions

The following table defines the primary symbols and variables used in the mathematical specifications and hypothesis tests.

Table 1: Symbols and Variable Dictionary.

Symbol	Definition
y_{isjr}	Observed moral judgment for participant i, scenario s, judged negotiator j, and role r.
y^*_{isjr}	Latent moral preference score (unbounded).
β_1	Regression coefficient representing the marginal effect of the predictor.
$\beta_{0.5}$	Median-regression coefficient from the non-parametric censored robustness model.
$Q_{0.5}(y^*_{isjr} \mid \mathbf{x}_{isjr})$	Conditional median of the latent bounded outcome given the predictors.
IRI_i	Empathy score (Average composite of the Interpersonal Reactivity Index).
CaseConfig_{isjr}	Judgment-level relational shorthand retained only for backward compatibility with older descriptive artifacts.
G_{isjr}	Judged negotiator relational status (ingroup, outgroup, or control) defined relative to the role-relevant reference actor.
A_{is}	Decision indicator distinguishing Accept from Reject.
C_{isjr}	Additional relational controls, including counterpart negotiator status, observer-side victim alignment, role, and demographic controls.
ICC	Intraclass Correlation: ratio of between-cluster variance to total variance.

Option 2: judgment-level relational modeling retains a victim x judged-negotiator shorthand for descriptive summaries, while H2 and H3 decompose relational structure into judged-negotiator status, decision outcome, judged-status x decision interactions, and additional relational controls for the counterpart negotiator and, in observer rows, victim alignment.

3 Hypotheses to Test

- **H1 (Empathy under relational controls):** Higher empathy predicts lower moral-judgment scores for harmful decisions after conditioning on judged-negotiator, counterpart, and observer-side victim relational controls.

- **H2 (Judged-status x decision contrasts):** Moral-judgment severity will differ significantly as a function of relational group membership, decision outcome, and their interaction.
- **H3 (Empathy x judged-status moderation):** The empathy effect may vary across judged-negotiator relational status, while decision outcome and relational controls remain in the model.

4 Mathematical Approach and Theoretical Foundations

The analysis employs two complementary estimators for bounded moral judgments. The primary specification is a Two-Limit Tobit model, which is theoretically appropriate for dependent variables that are strictly bounded within a known interval. In this context, negotiator-specific moral judgments (y_{isjr}) are observed on a scale from -9 to 9. The Tobit model assumes the existence of a latent, unobserved preference index (y^*_{isjr}) that follows a linear relationship. Option 2 replaces isolated ingroup/outgroup indicators with explicit judgment-level relational variables built from the paired-group structure of each judgment. The analytic hypothesis models decompose that structure into the judged negotiator's relational status, decision outcome (Accept/Reject), their interaction, and additional relational controls for the counterpart negotiator and, in observer rows, victim alignment. H1 retains accepted-sample victim x judged-negotiator case contrasts, while H2 and H3 use judged-negotiator status, decision outcome, their interaction, and additional relational controls:

$$y^*_{isjr} = \beta_0 + \beta_1 \text{Empathy}_i + \boldsymbol{\beta}'_2 G_{isjr} + \beta_3 A_{is} + \boldsymbol{\beta}'_4 (G_{isjr} \times A_{is}) + \boldsymbol{\beta}'_5 C_{isjr} + \epsilon_{isjr}, \quad \epsilon_{isjr} \sim N(0, \sigma^2)$$

The actual observed judgment y_{isjr} relates to this latent variable via the censoring transformation:

$$y_{isjr} = \max(-9, \min(9, y^*_{isjr}))$$

This approach prevents the 'ceiling' and 'floor' effects from biasing the linear coefficients, as would occur in standard OLS regression.

Because the Tobit model relies on a Gaussian latent-error assumption, the pipeline also fits a distribution-robust censored median specification implemented as interval-censored quantile regression ($p = 0.5$). This complementary estimator targets the conditional median of the latent bounded outcome and is less sensitive to heavy tails and non-normal disturbances:

$$Q_{0.5}(y^*_{isjr} \mid \mathbf{x}_{isjr}) = \mathbf{x}'_{isjr} \beta_{0.5}$$

The non-parametric branch preserves the censoring structure while relaxing the parametric normality assumption. The Tobit and robustness results should therefore be interpreted jointly: Tobit provides the clustered parametric benchmark with participant-clustered standard errors by id, while the non-parametric branch first establishes a converged full-sample censored median fit and then, in the default pipeline, adds participant-level cluster bootstrap inference that resamples ids with replacement and retains all repeated observations from each sampled participant. In both branches, id is used only to account for within-participant dependence and is not a substantive explanatory variable.

5 Analysis of Sample Size Impact

The statistical validity of our inferences depends on the trade-off between False Positives (Type I error, α) and False Negatives (Type II error, β). Given the clustered nature of our data, sample size impact is not merely a count of observations, but a function of cluster correlation.

5.0.1 Step-by-Step Sensitivity Analysis

To determine the impact of our sample size on our ability to detect effects, we follow these steps:

1. Calculate the Design Effect (Deff): As multiple judgments are nested within individuals, we adjust for the Intraclass Correlation (ICC).

$$Deff = 1 + (m - 1) \times ICC$$

where m is the average number of scenarios per participant.

2. Determine the Effective Sample Size (ESS): The ESS represents the number of independent observations that would provide the same statistical power as our clustered sample.

$$ESS = \frac{n_{total}}{Deff}$$

3. Inference Impact: A higher ICC reduces the ESS, thereby increasing the Standard Error of our Tobit coefficients. If the ESS is low, our models become 'conservative', increasing the risk of Type II errors (failing to support a hypothesis). By using clustered robust standard errors for Tobit and participant-level cluster bootstrap inference for the non-parametric robustness model after each converged full-sample fit, we ensure that both sets of p-values acknowledge this reduced information density, protecting the integrity of our Type I error threshold ($\alpha = 0.05$).

Based on the current run, the observed Intraclass Correlation (ICC) is 0.717, with an Effective Sample Size (ESS) of 266.9.

6 Descriptive Statistics

The empathy profile of the sample is visualized in Figure 1, showing the average scores across the four IRI latent variables.

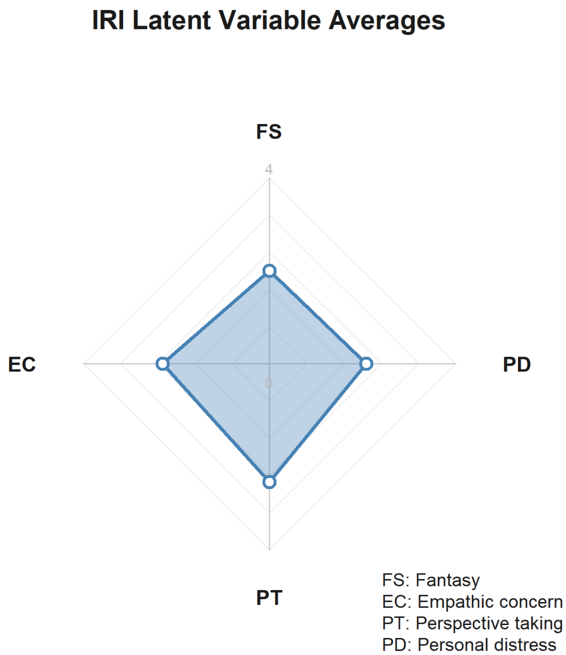

Figure 1: IRI Latent Variable Averages (Radar Plot profile).

The descriptive branch no longer centers the main report on victim x negotiator case-label tables; the hypothesis tests below work directly with negotiator-level relational predictors.

7 Bi-variate Statistics

The correlation matrix between the psychometric subscales and the mean moral judgment is presented below.

Table 2: Correlation Matrix: IRI Subscales and Moral Judgment.

Fantasy	Empathic Concern	Perspective Taking	Personal Distress	Total IRI	Mean Judgment
1.000	0.500	0.554	0.509	0.830	-0.030
0.500	1.000	0.424	0.649	0.818	-0.116
0.554	0.424	1.000	0.301	0.730	0.074
0.509	0.649	0.301	1.000	0.762	-0.010
0.830	0.818	0.730	0.762	1.000	-0.029
-0.030	-0.116	0.074	-0.010	-0.029	1.000

Visual representations of these relationships are provided in Figure 2.

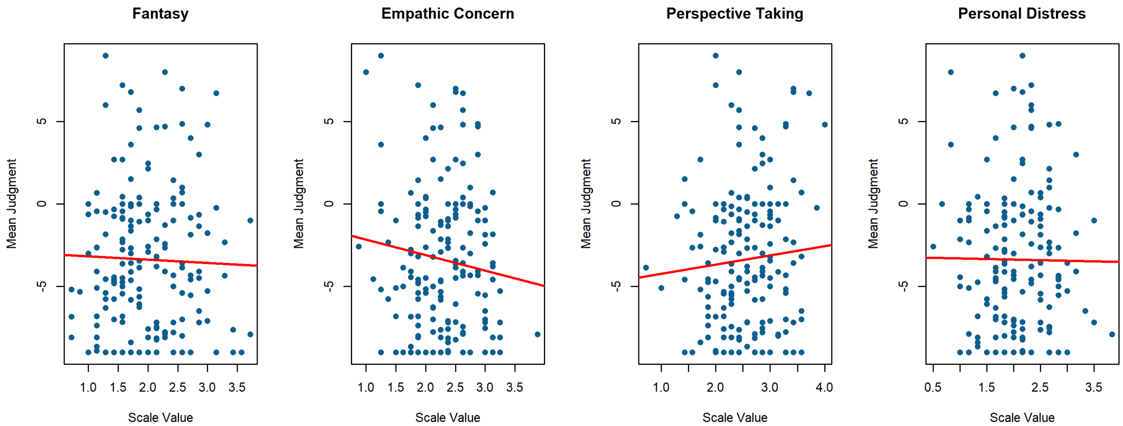

Figure 2: Bivariate Scatters: IRI Scales vs. Mean Judgment.

8 Estimator Fit Summary

The following table consolidates the fit-status information for the primary Tobit models and the non-parametric robustness branch. In the default pipeline, participant-level cluster bootstrap launches immediately after a converged full-sample non-parametric fit is available; if bootstrap is disabled manually or too few bootstrap refits converge, that status is shown explicitly. This run uses 5 participant-level bootstrap replicates for the non-parametric branch. A bootstrap_failed status means the full-sample CLAD fit converged but none of the attempted bootstrap refits converged; it does not imply that the full non-parametric estimator itself failed to converge.

Table 3: Estimator fit summary across Tobit and cluster-aware non-parametric specifications.

Model	Approach	Status	Converged	Iterations	BootstrapReplicates	BootstrapSuccessful	Observations	Participants	ClusterUnit
H1_A_Total_CLAD	CLAD	not_converged	FALSE	2000	5	NA	1939	199	id
H1_B_Constructs_CLAD	CLAD	not_converged	FALSE	2000	5	NA	1939	199	id
H2a_A_Total_CLAD	CLAD	not_converged	FALSE	2000	5	NA	1678	198	id
H2a_B_Constructs_CLAD	CLAD	not_converged	FALSE	2000	5	NA	1678	198	id
H2b_A_Total_CLAD	CLAD	not_converged	FALSE	2000	5	NA	3980	199	id
H2b_B_Constructs_CLAD	CLAD	not_converged	FALSE	2000	5	NA	3980	199	id
H3_A_Total_CLAD	CLAD	not_converged	FALSE	2000	5	NA	3980	199	id
H3_B_Constructs_CLAD	CLAD	not_converged	FALSE	2000	5	NA	3980	199	id
H1_A_Total_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H1_B_Constructs_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H2a_A_Total_Tobit	Tobit	completed	NA	6	NA	NA	1678	198	id
H2a_B_Constructs_Tobit	Tobit	completed	NA	6	NA	NA	1678	198	id
H2b_A_Total_Tobit	Tobit	completed	NA	6	NA	NA	3980	199	id
H2b_B_Constructs_Tobit	Tobit	completed	NA	6	NA	NA	3980	199	id
H3_A_Total_Tobit	Tobit	completed	NA	6	NA	NA	3980	199	id
H3_B_Constructs_Tobit	Tobit	completed	NA	6	NA	NA	3980	199	id

8.1 Hypothesis Significance Summary

The following concise table lists only hypothesis-relevant predictors that reached at least $p < 0.10$, using conventional symbols to indicate strength of evidence (+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). If the non-parametric bootstrap is disabled, too sparse, or the censored median fit does not converge, the non-parametric column reports that status instead of inferential symbols. Dynamic figures are generated only for predictors that appear in this table with at least one significance symbol.

Table 4: Hypothesis-level significance summary across Tobit and cluster-aware non-parametric models.

Hypothesis	Tobit support	Non-parametric support
H1	Empathy: Empathic concern*; Empathy: Perspective taking*	None
H2	None	None
H3	Personal distress x judged negotiator outgroup**; Empathic concern x judged negotiator outgroup*	None

8.2 Significance-Driven Figures

Only hypothesis-relevant predictors that reach $p < 0.10$ or better are visualized automatically. These dynamic figures use the saved Tobit and clustered non-parametric outputs, and participant id remains only an inference-level clustering unit rather than a substantive explanatory variable.

H1: Empathy under relational controls Empathy: Empathic concern is statistically significant in the Tobit model (*, $p = 0.023$). Figure 3 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted latent judgment.

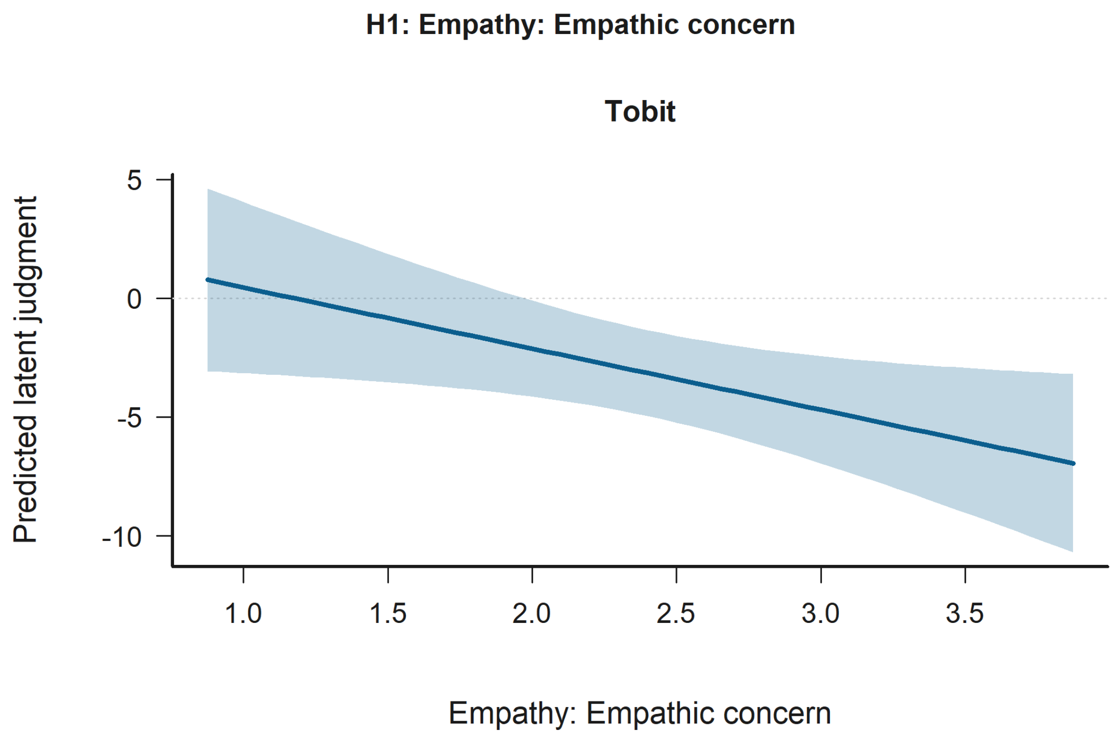

Figure 3: Effect Plot for Empathy: Empathic concern in H1: Empathy under relational controls. Support comes from the Tobit model (*, p = 0.023). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Perspective taking is statistically significant in the Tobit model (*, p = 0.045). Figure 4 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted latent judgment.

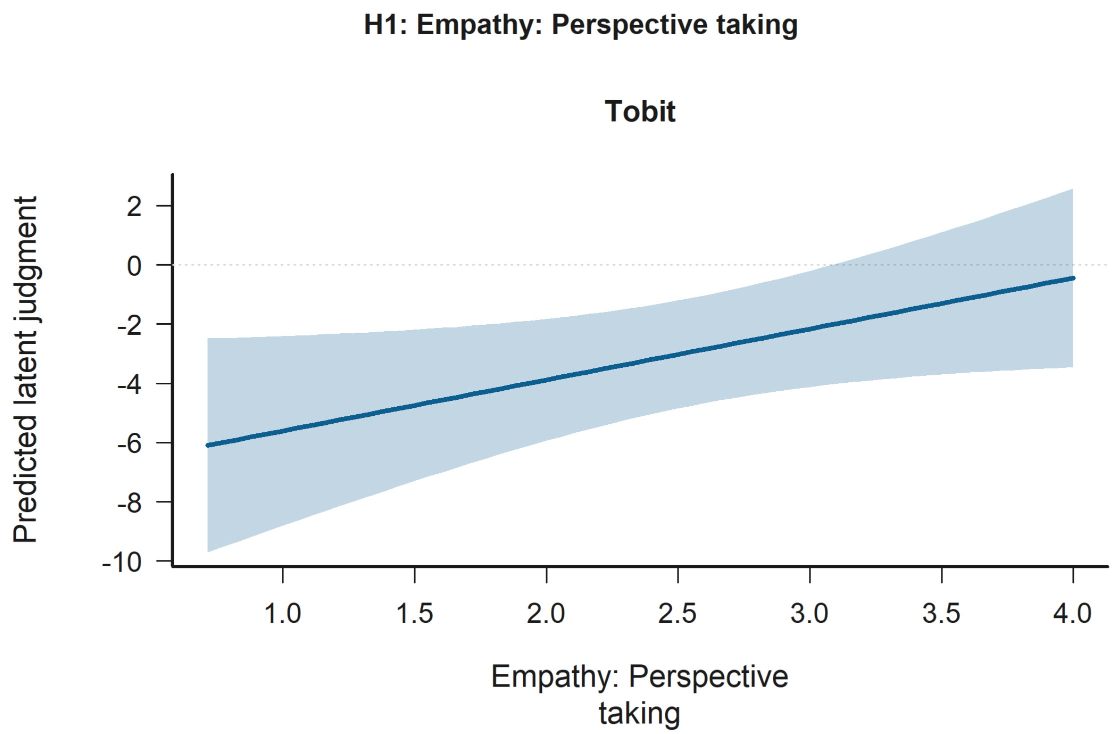

Figure 4: Effect Plot for Empathy: Perspective taking in H1: Empathy under relational controls. Support comes from the Tobit model (*, p = 0.045). The panels show predicted latent judgments with 95% confidence intervals.

H3: Empathy x judged-status moderation Personal distress x judged negotiator outgroup is statistically significant in the Tobit model (**, p = 0.002). Figure 5 shows that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is Judged negotiator outgroup.

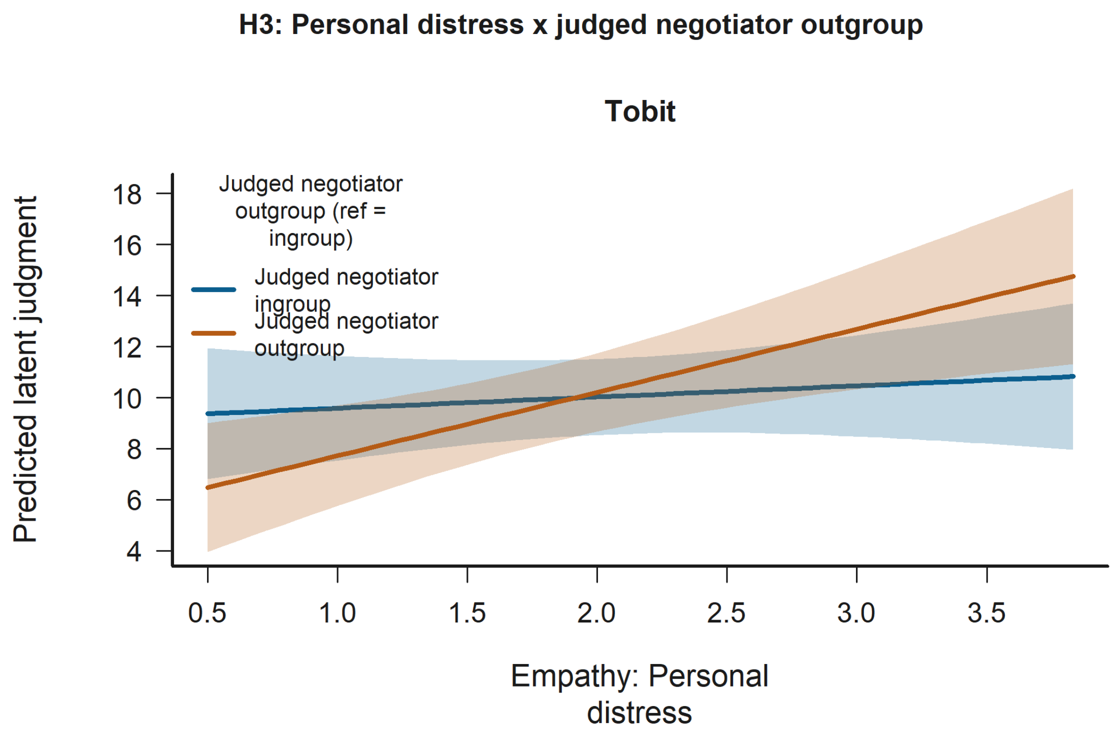

Figure 5: Interaction Plot for Personal distress x judged negotiator outgroup in H3: Empathy x judged-status moderation. Support comes from the Tobit model (**, p = 0.002). The panels show predicted latent judgments with 95% confidence intervals.

Empathic concern x judged negotiator outgroup is statistically significant in the Tobit model (*, p = 0.025). Figure 6 shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is Judged negotiator outgroup.

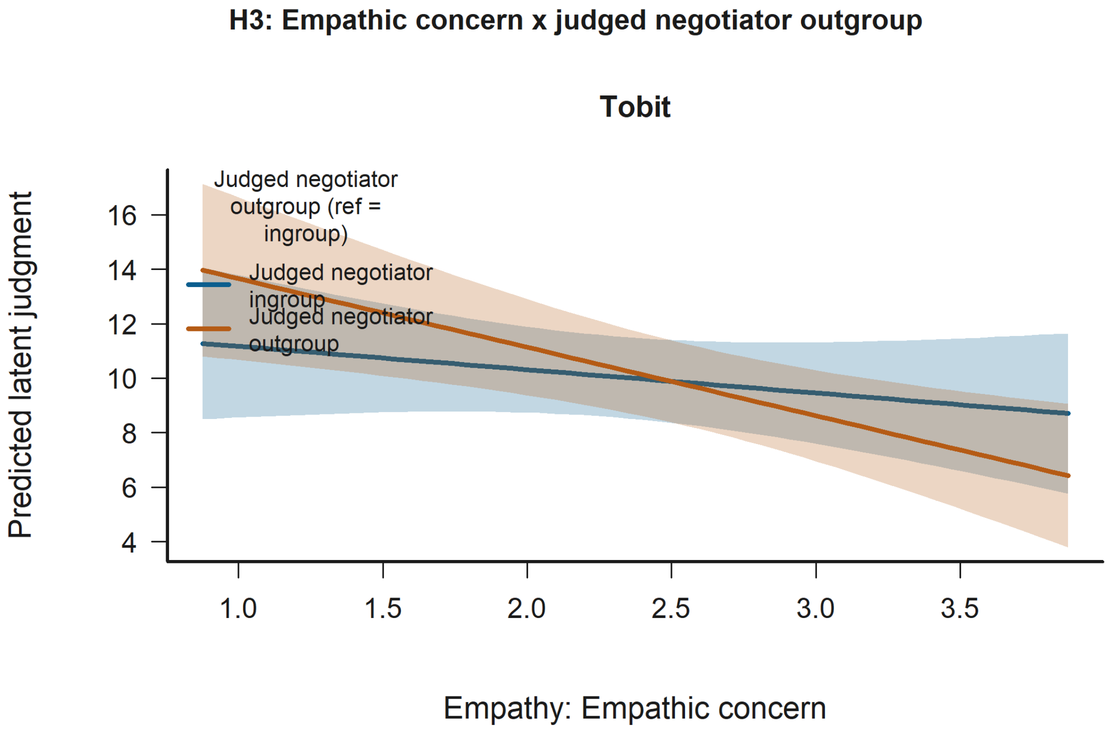

Figure 6: Interaction Plot for Empathic concern x judged negotiator outgroup in H3: Empathy x judged-status moderation. Support comes from the Tobit model (*, p = 0.025). The panels show predicted latent judgments with 95% confidence intervals.

8.3 All Significant Predictors ($p < 0.10$)

The following figures extend beyond the hypothesis-target terms and visualize every predictor that reaches $p < 0.10$ in the available H1-H3 Tobit or clustered non-parametric models. This includes significant controls such as age when they clear the threshold.

Accepted-decision sample Observer role (ref = victim) is statistically significant in:

- H1: Empathy under relational controls Model B (Tobit)
- H1: Empathy under relational controls Model A (Tobit)

Figure 7 shows that predicted latent judgment is higher for Observer role than for Victim role.

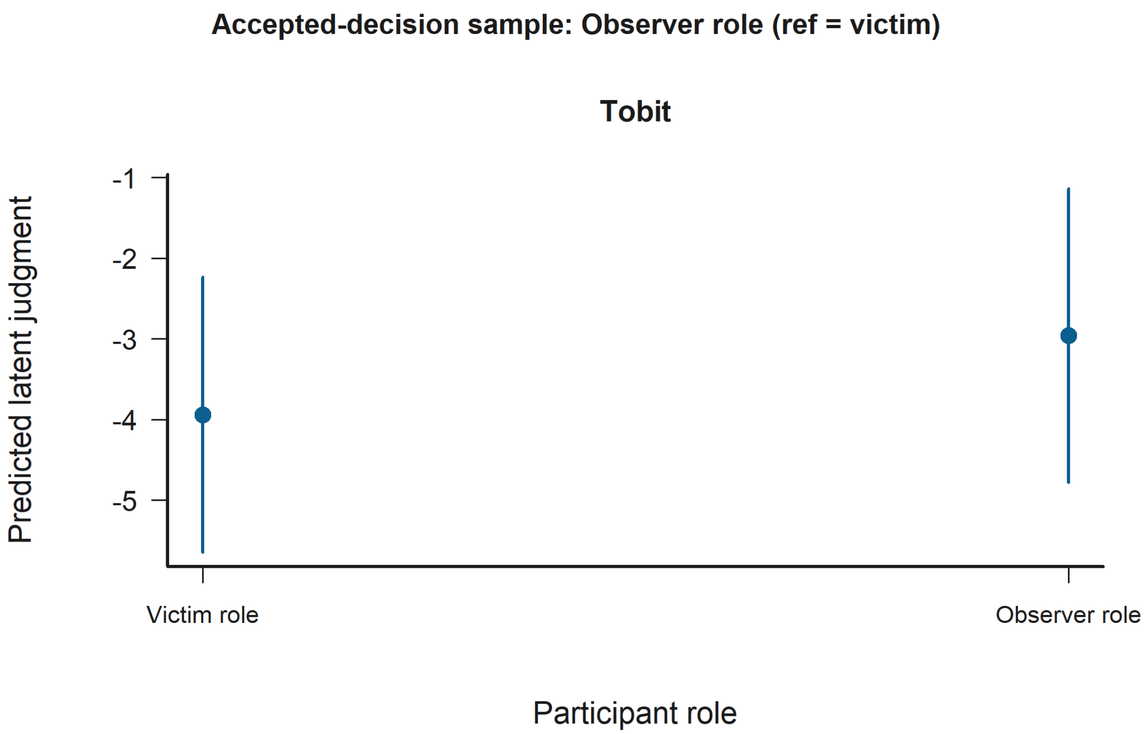

Figure 7: Grouped Prediction Plot for Observer role (ref = victim) in the Accepted-decision sample. Statistically significant support appears in H1: Empathy under relational controls Model B (Tobit) and H1: Empathy under relational controls Model A (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Empathic concern is statistically significant in:

- H1: Empathy under relational controls Model B (Tobit)

Figure 8 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted latent judgment.

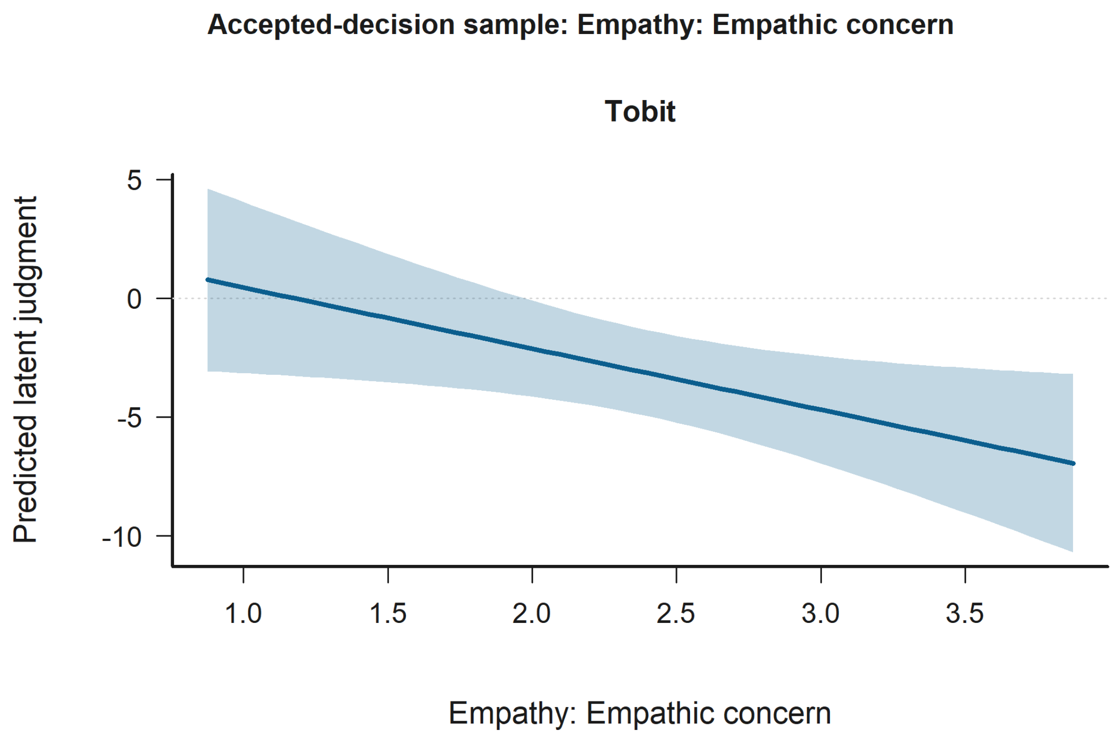

Figure 8: Effect Plot for Empathy: Empathic concern in the Accepted-decision sample. Statistically significant support appears in H1: Empathy under relational controls Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Age is statistically significant in:

- H1: Empathy under relational controls Model A (Tobit)
- H1: Empathy under relational controls Model B (Tobit)

Figure 9 shows that across the observed range, higher Age corresponds to higher predicted latent judgment.

Figure 9: Effect Plot for Age in the Accepted-decision sample. Statistically significant support appears in H1: Empathy under relational controls Model A (Tobit) and H1: Empathy under relational controls Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Perspective taking is statistically significant in:

- H1: Empathy under relational controls Model B (Tobit)

Figure 10 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted latent judgment.

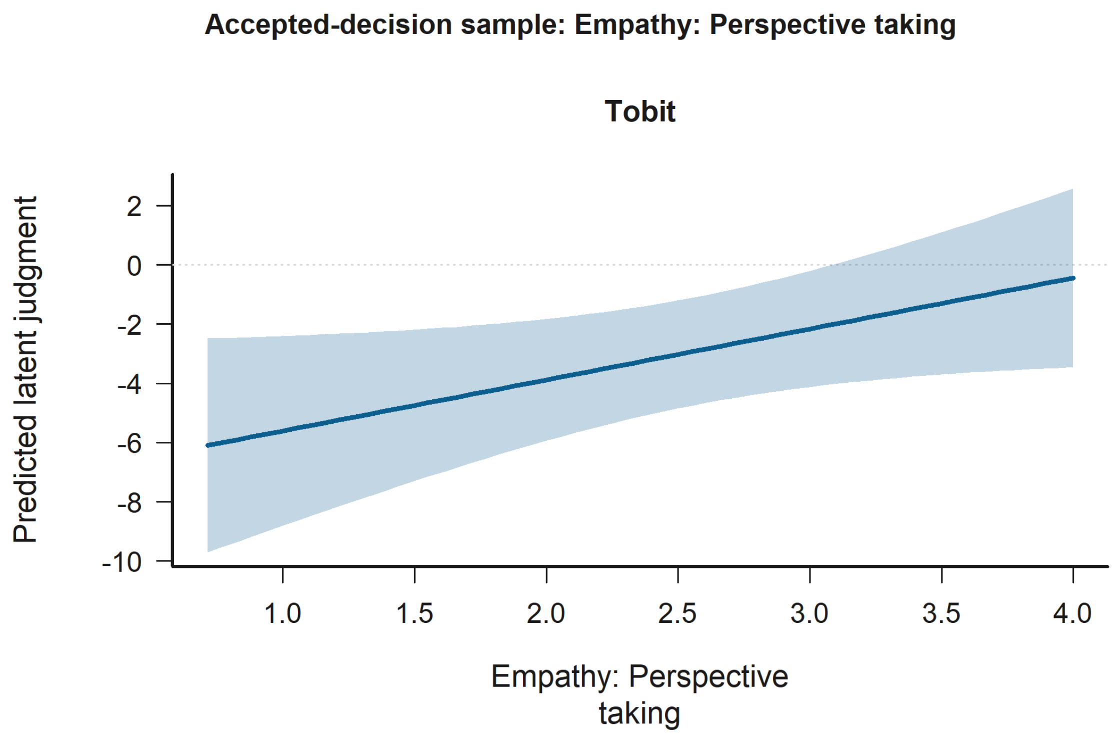

Figure 10: Effect Plot for Empathy: Perspective taking in the Accepted-decision sample. Statistically significant support appears in H1: Empathy under relational controls Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Engineering participant (ref = humanities) is statistically significant in:

- H1: Empathy under relational controls Model A (Tobit)
- H1: Empathy under relational controls Model B (Tobit)

Figure 11 shows that predicted latent judgment is higher for Engineering participant than for Humanities participant.

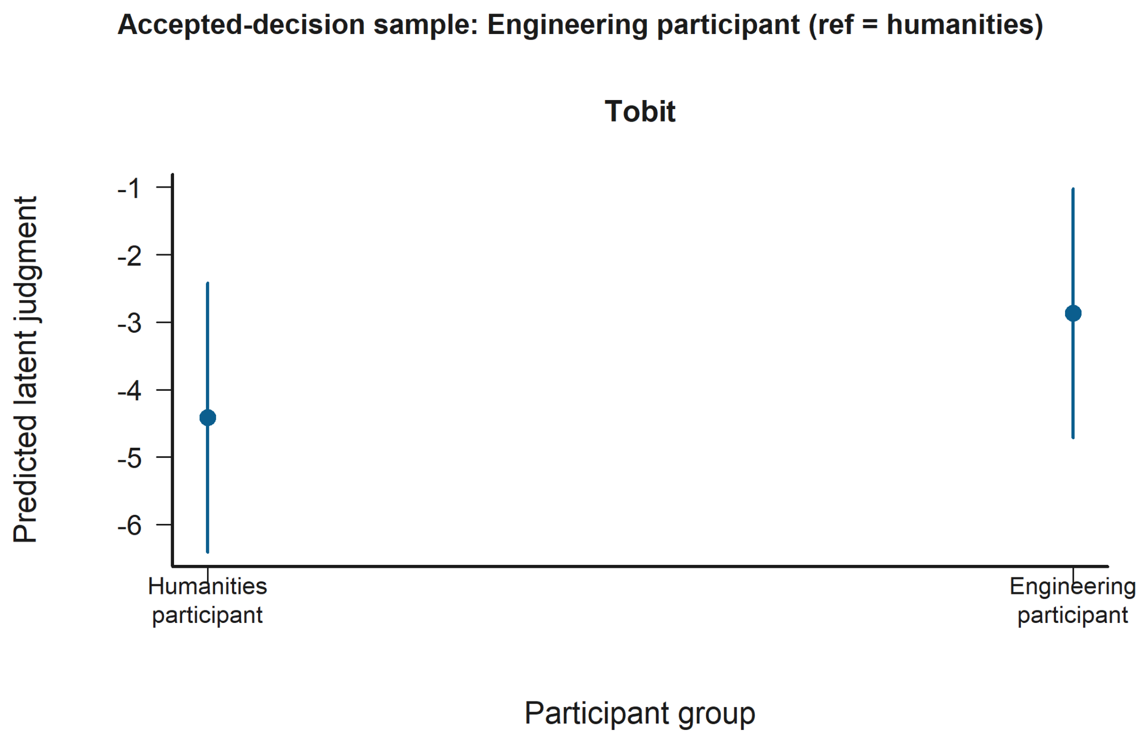

Figure 11: Grouped Prediction Plot for Engineering participant (ref = humanities) in the Accepted-decision sample. Statistically significant support appears in H1: Empathy under relational controls Model A (Tobit) and H1: Empathy under relational controls Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Betrayal sample Negotiator accepted harmful deal is statistically significant in:

- H2: Judged-status x decision contrasts Model A (Tobit)
- H2: Judged-status x decision contrasts Model B (Tobit)

Figure 12 shows that predicted latent judgment is lower for Accept harmful deal than for Reject harmful deal.

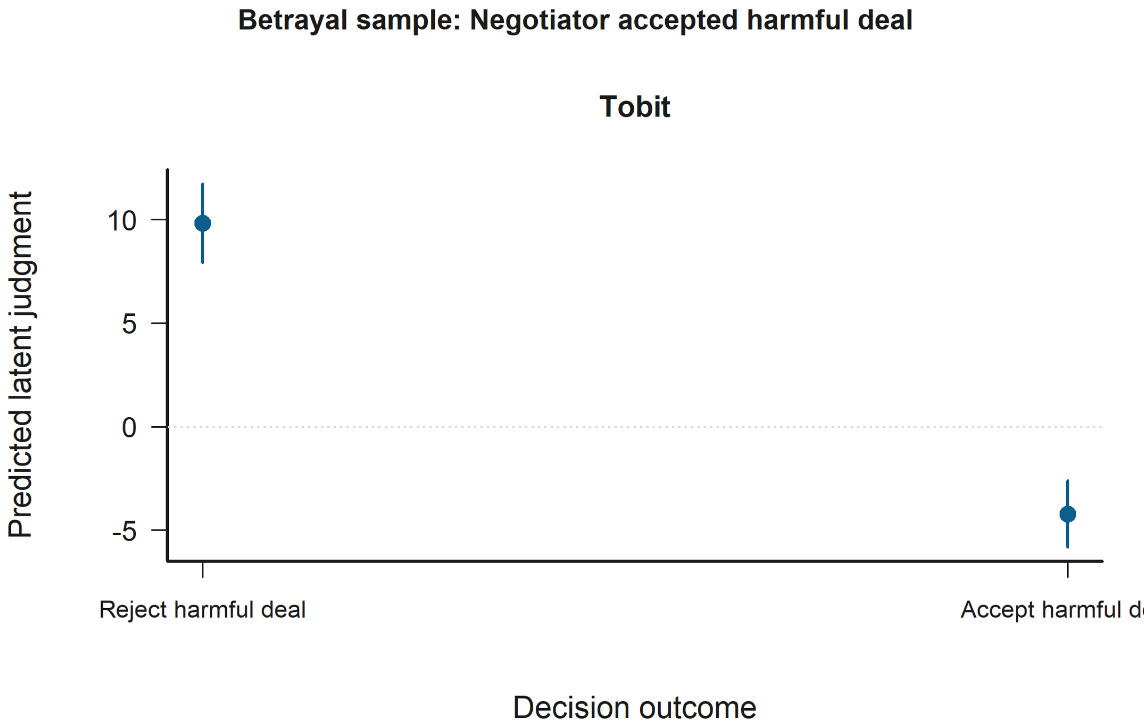

Figure 12: Grouped Prediction Plot for Negotiator accepted harmful deal in the Betrayal sample. Statistically significant support appears in H2: Judged-status x decision contrasts Model A (Tobit) and H2: Judged-status x decision contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Perspective taking is statistically significant in:

- H2: Judged-status x decision contrasts Model B (Tobit)

Figure 13 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted latent judgment.

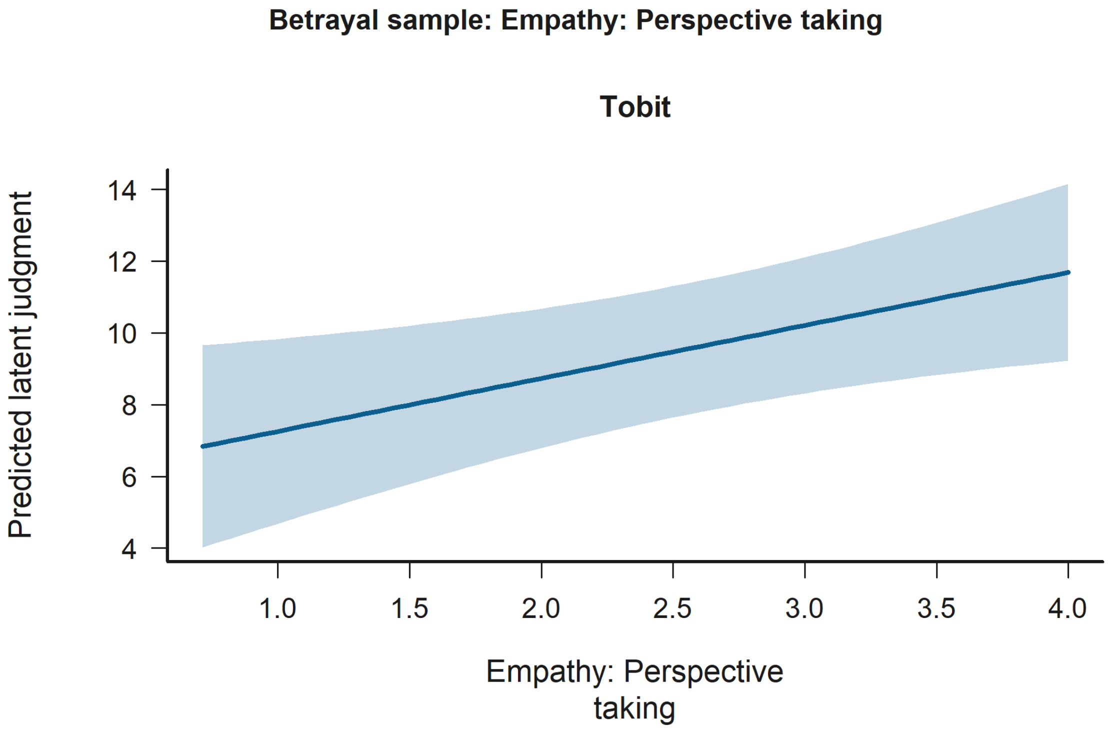

Figure 13: Effect Plot for Empathy: Perspective taking in the Betrayal sample. Statistically significant support appears in H2: Judged-status x decision contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Socioeconomic status is statistically significant in:

- H2: Judged-status x decision contrasts Model A (Tobit)
- H2: Judged-status x decision contrasts Model B (Tobit)

Figure 14 shows that predicted latent judgment is higher for Socioeconomic status 5 than for Socioeconomic status 0.

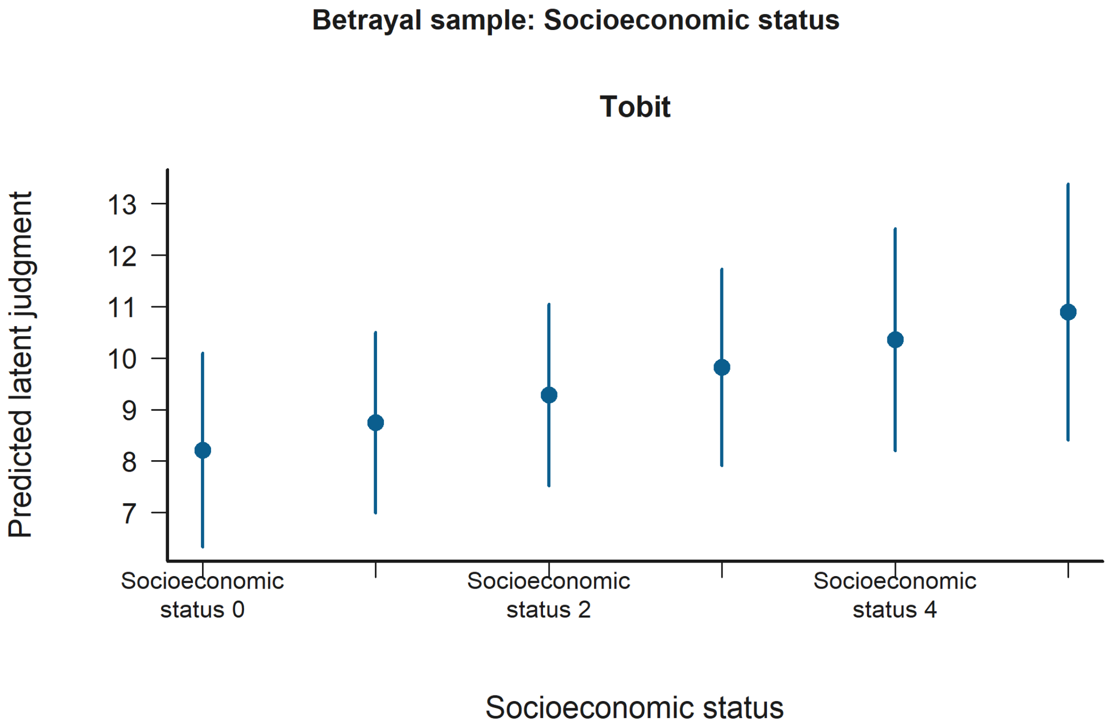

Figure 14: Grouped Prediction Plot for Socioeconomic status in the Betrayal sample. Statistically significant support appears in H2: Judged-status x decision contrasts Model A (Tobit) and H2: Judged-status x decision contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Empathic concern is statistically significant in:

- H2: Judged-status x decision contrasts Model B (Tobit)

Figure 15 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted latent judgment.

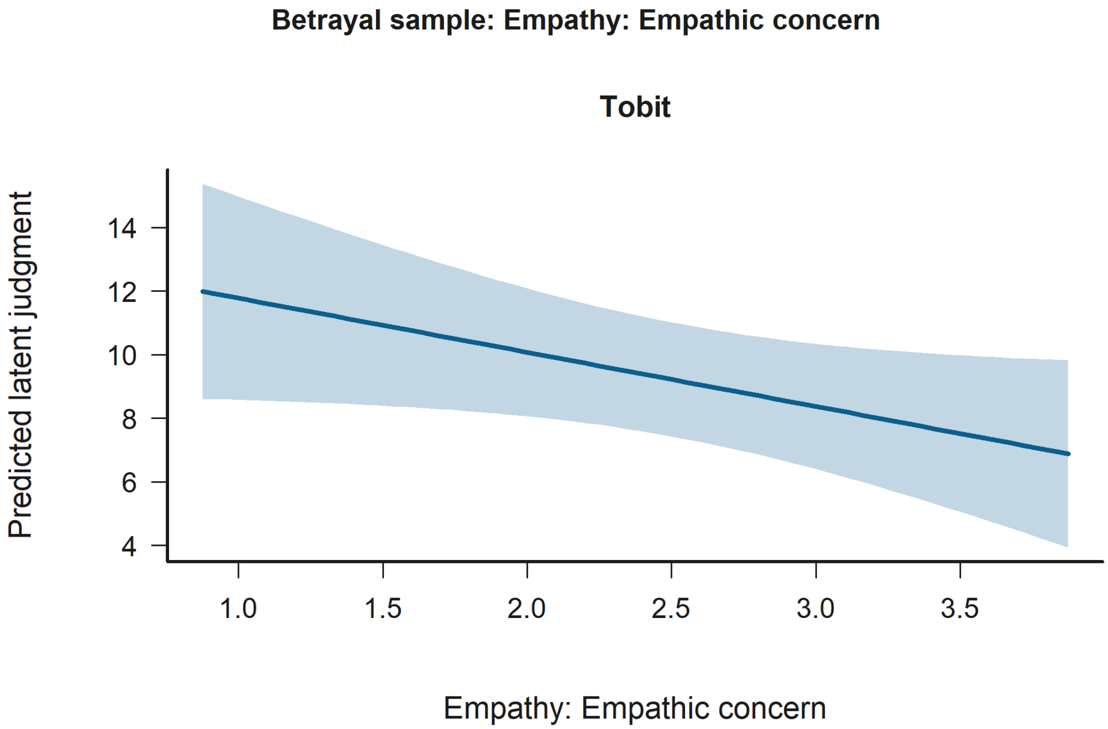

Figure 15: Effect Plot for Empathy: Empathic concern in the Betrayal sample. Statistically significant support appears in H2: Judged-status x decision contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Full judgment sample Negotiator accepted harmful deal is statistically significant in:

- H2: Judged-status x decision contrasts Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H2: Judged-status x decision contrasts Model A (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 16 shows that predicted latent judgment is lower for Accept harmful deal than for Reject harmful deal.

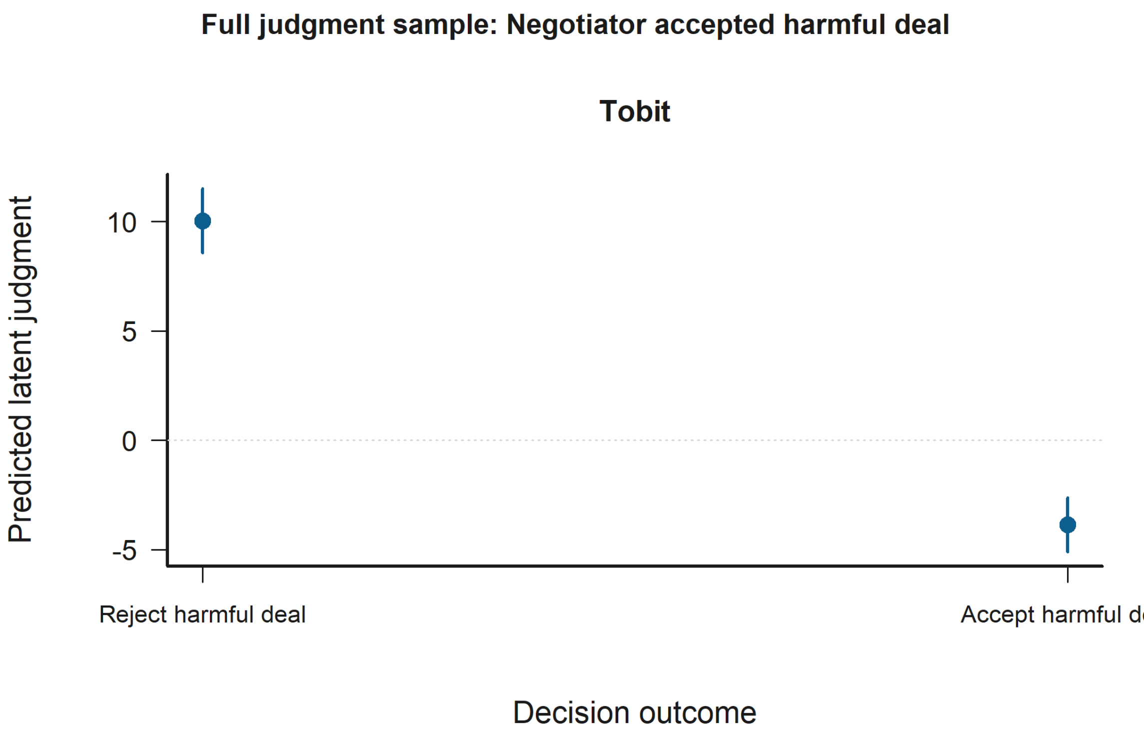

Figure 16: Grouped Prediction Plot for Negotiator accepted harmful deal in the Full judgment sample. Statistically significant support appears in H2: Judged-status x decision contrasts Model B (Tobit), H3: Empathy x judged-status moderation Model B (Tobit), H2: Judged-status x decision contrasts Model A (Tobit), and H3: Empathy x judged-status moderation Model A (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Personal distress x judged negotiator outgroup is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 17 shows that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is Judged negotiator outgroup.

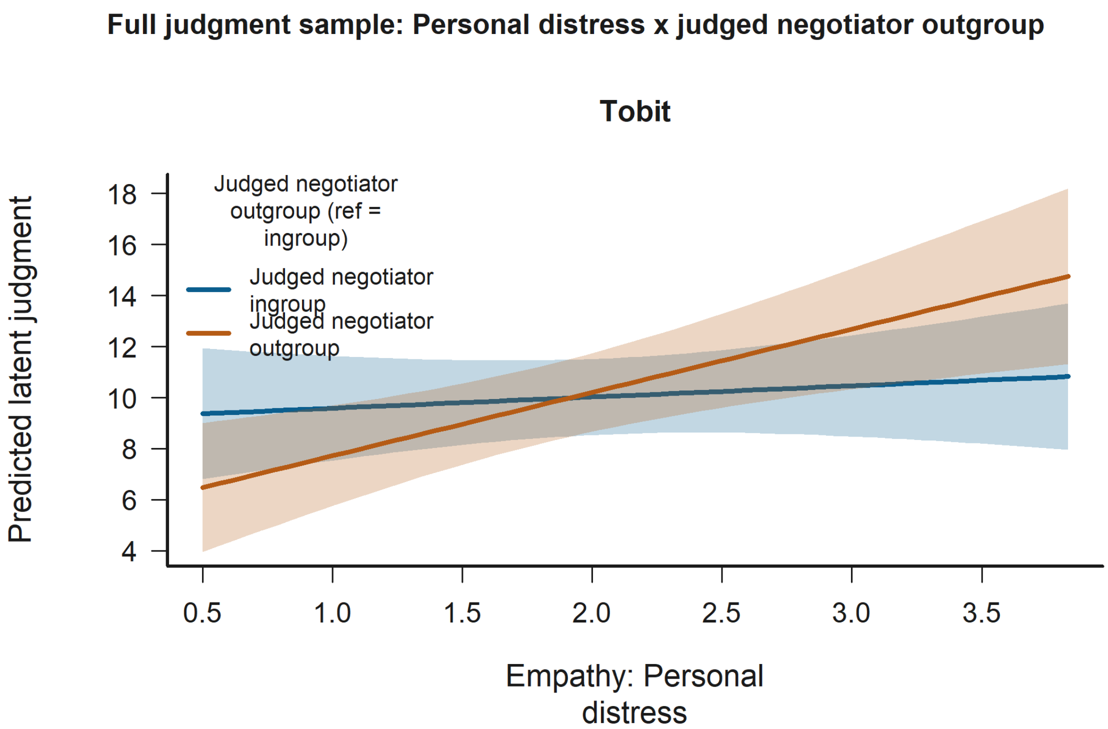

Figure 17: Interaction Plot for Personal distress x judged negotiator outgroup in the Full judgment sample. Statistically significant support appears in H3: Empathy x judged-status moderation Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Perspective taking is statistically significant in:

- H2: Judged-status x decision contrasts Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 18 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted latent judgment.

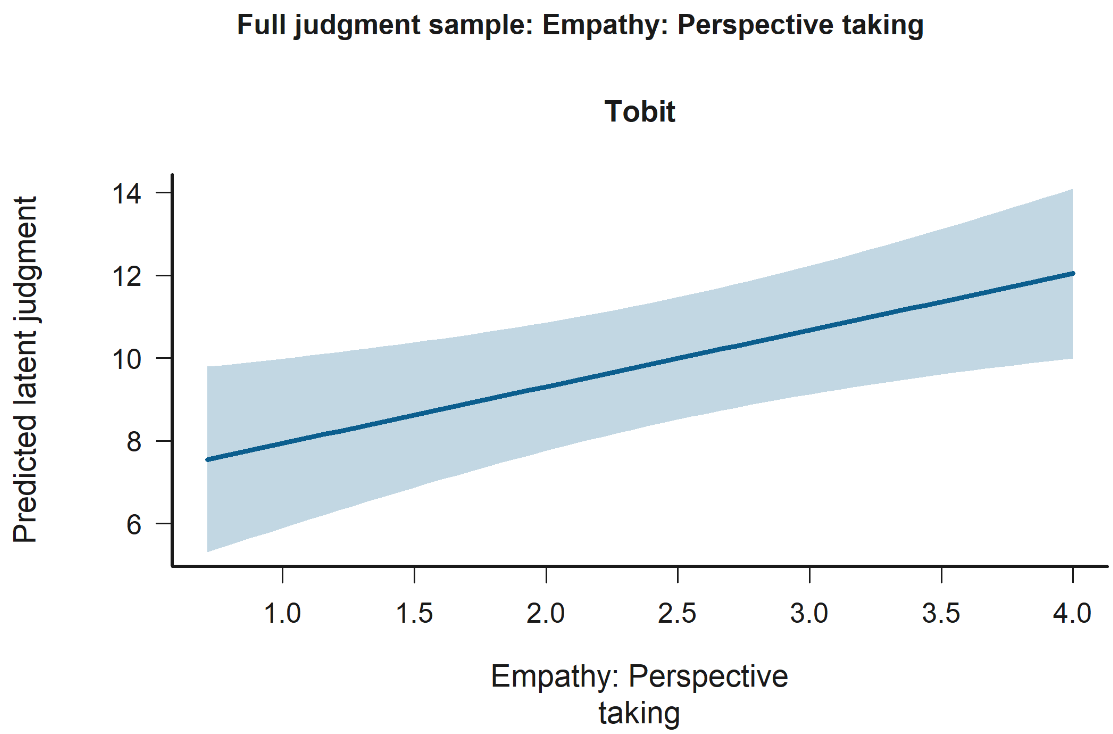

Figure 18: Effect Plot for Empathy: Perspective taking in the Full judgment sample. Statistically significant support appears in H2: Judged-status x decision contrasts Model B (Tobit) and H3: Empathy x judged-status moderation Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Engineering participant (ref = humanities) is statistically significant in:

- H2: Judged-status x decision contrasts Model A (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H2: Judged-status x decision contrasts Model B (Tobit)

Figure 19 shows that predicted latent judgment is higher for Engineering participant than for Humanities participant.

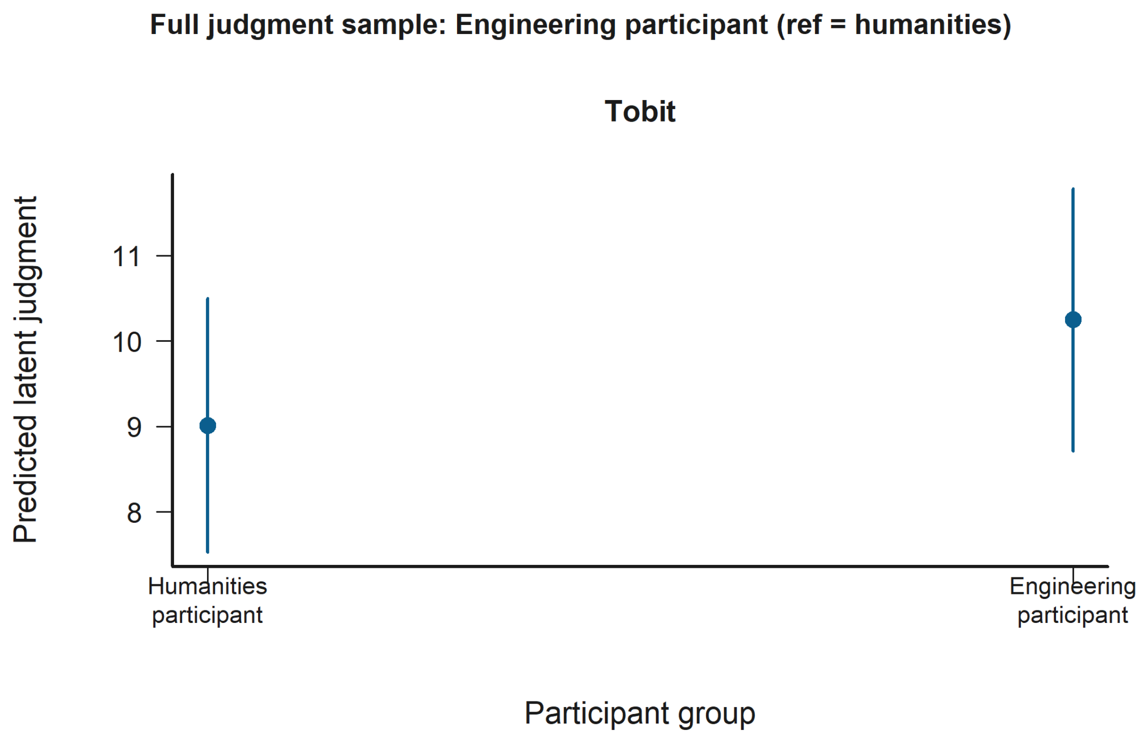

Figure 19: Grouped Prediction Plot for Engineering participant (ref = humanities) in the Full judgment sample. Statistically significant support appears in H2: Judged-status x decision contrasts Model A (Tobit), H3: Empathy x judged-status moderation Model A (Tobit), H3: Empathy x judged-status moderation Model B (Tobit), and H2: Judged-status x decision contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathic concern x judged negotiator outgroup is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 20 shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is Judged negotiator outgroup.

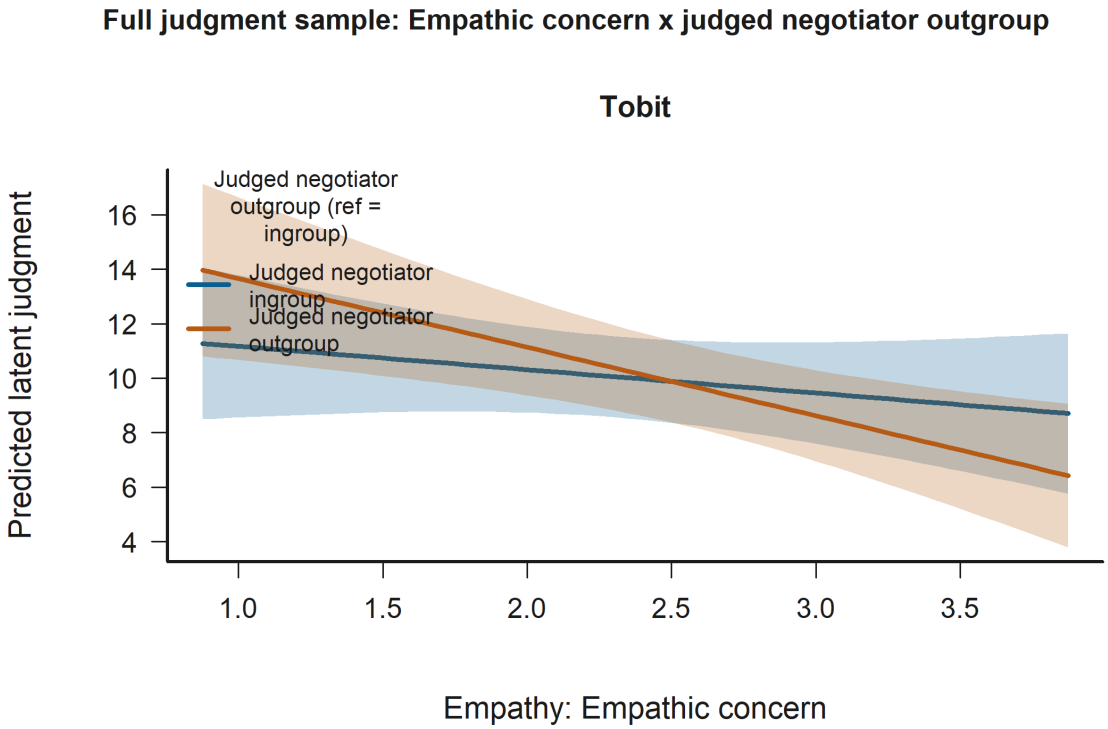

Figure 20: Interaction Plot for Empathic concern x judged negotiator outgroup in the Full judgment sample. Statistically significant support appears in H3: Empathy x judged-status moderation Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Empathic concern is statistically significant in:

- H2: Judged-status x decision contrasts Model B (Tobit)

Figure 21 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted latent judgment.

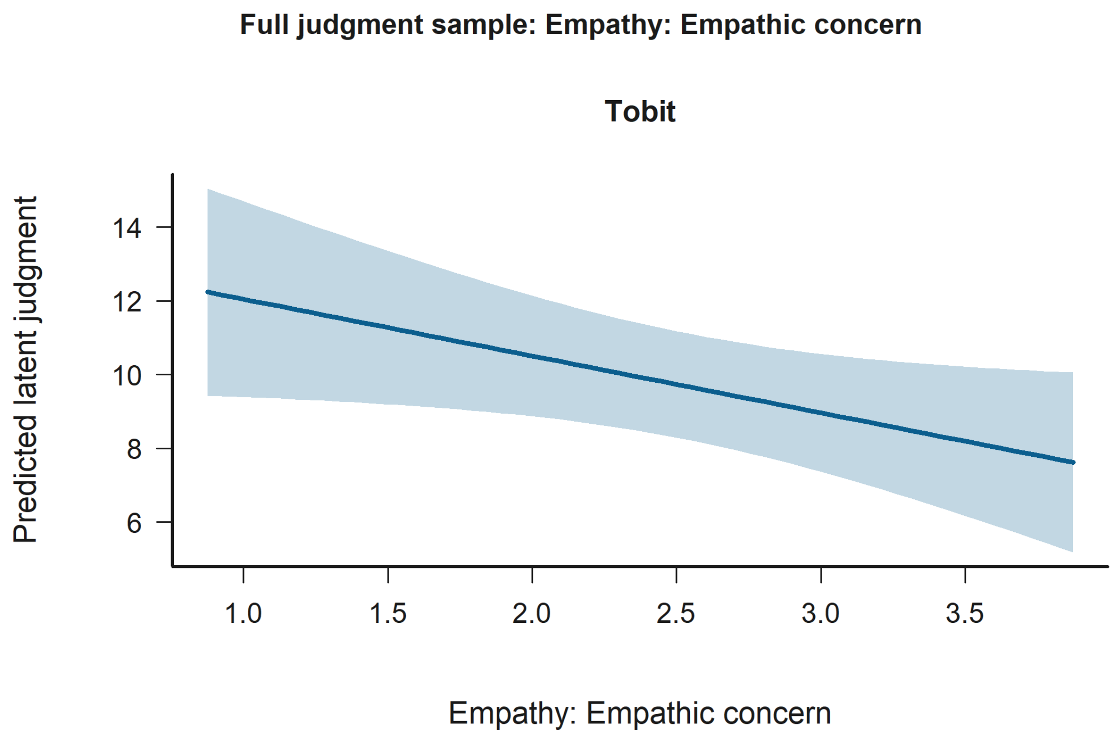

Figure 21: Effect Plot for Empathy: Empathic concern in the Full judgment sample. Statistically significant support appears in H2: Judged-status x decision contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Socioeconomic status is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)
- H2: Judged-status x decision contrasts Model A (Tobit)

Figure 22 shows that predicted latent judgment is higher for Socioeconomic status 5 than for Socioeconomic status 0.

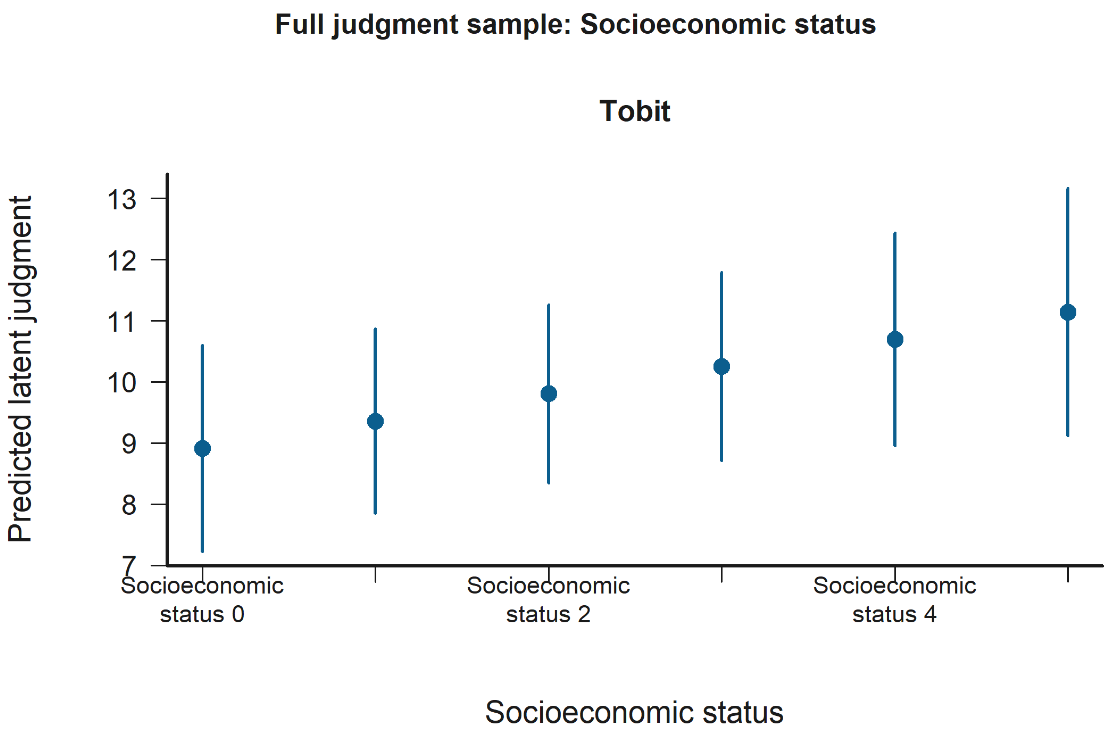

Figure 22: Grouped Prediction Plot for Socioeconomic status in the Full judgment sample. Statistically significant support appears in H3: Empathy x judged-status moderation Model A (Tobit) and H2: Judged-status x decision contrasts Model A (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Judged negotiator outgroup (ref = ingroup) is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 23 shows that predicted latent judgment is higher for Judged negotiator outgroup than for Judged negotiator ingroup.

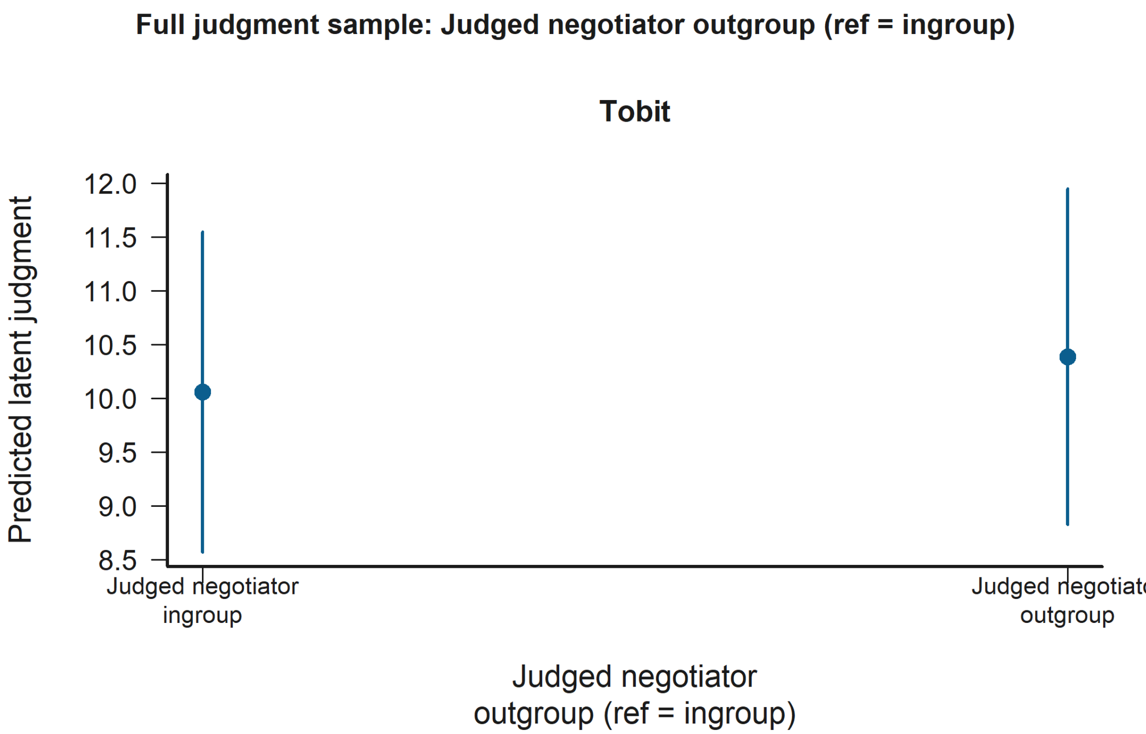

Figure 23: Grouped Prediction Plot for Judged negotiator outgroup (ref = ingroup) in the Full judgment sample. Statistically significant support appears in H3: Empathy x judged-status moderation Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

Empathy: Personal distress is statistically significant in:

- H2: Judged-status x decision contrasts Model B (Tobit)

Figure 24 shows that across the observed range, higher Empathy: Personal distress corresponds to higher predicted latent judgment.

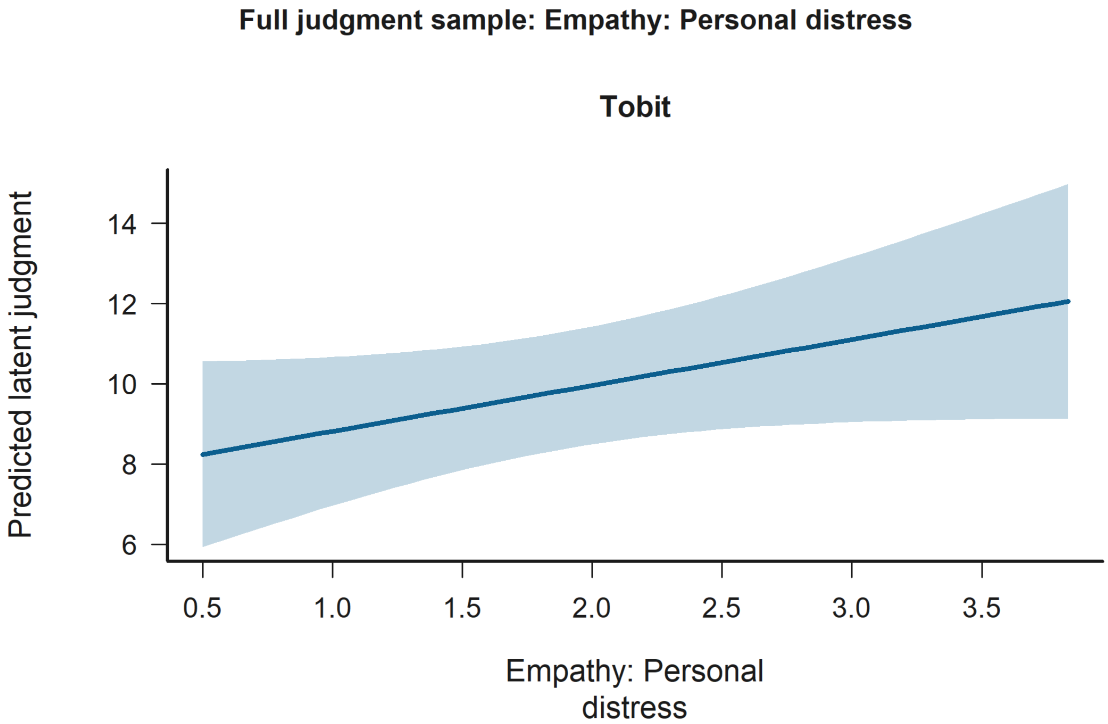

Figure 24: Effect Plot for Empathy: Personal distress in the Full judgment sample. Statistically significant support appears in H2: Judged-status x decision contrasts Model B (Tobit). The panels show predicted latent judgments with 95% confidence intervals.

8.4 Integrated Hypothesis Conclusions

The following summary restates each original hypothesis and indicates whether the available Tobit estimates and the cluster-aware non-parametric models support it in the current data. Non-parametric conclusions are drawn when the participant-level bootstrap inference is available and are otherwise labeled explicitly.

- H1. Original hypothesis: Higher empathy predicts lower moral-judgment scores for harmful decisions after conditioning on judged-negotiator, counterpart, and observer-side victim relational controls. Tobit conclusion: the evidence is mixed but offers partial support for the hypothesis. Model A does not support the hypothesis; Empathy composite (average) is negative but not statistically significant ($p = 0.492$). Model B supports the hypothesis through Empathy: Empathic concern with a negative association ($p = 0.023$). Additional statistically significant signals include Observer role (ref = victim) with a positive association ($p = 0.023$) and Age with a positive association ($p = 0.034$). Non-parametric conclusion: no converged second-phase non-parametric model is available, so the robustness check is inconclusive for this hypothesis.
- H2a. Original hypothesis: Moral-judgment severity should vary with the judged negotiator's ingroup-versus-outgroup status, and that relational effect should depend on whether the

harmful deal was accepted or rejected. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; none of the judged-negotiator relational-status terms and their decision interaction in the non-control sample are statistically significant, and the closest signal is Judged negotiator outgroup (ref = ingroup) with a negative association ($p = 0.706$). Model B does not support the hypothesis; none of the judged-negotiator relational-status terms and their decision interaction in the non-control sample are statistically significant, and the closest signal is Accepted harmful deal x judged negotiator outgroup with a negative association ($p = 0.896$). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Empathy: Perspective taking with a positive association ($p = 0.012$). Non-parametric conclusion: no converged second-phase non-parametric model is available, so the robustness check is inconclusive for this hypothesis.

- H2b. Original hypothesis: Moral-judgment severity should vary with the judged negotiator's ingroup-versus-outgroup-versus-control status, and that relational effect should depend on whether the harmful deal was accepted or rejected. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; none of the judged-negotiator relational-status terms and their decision interaction in the full sample are statistically significant, and the closest signal is Accepted harmful deal x judged negotiator outgroup with a negative association ($p = 0.396$). Model B does not support the hypothesis; none of the judged-negotiator relational-status terms and their decision interaction in the full sample are statistically significant, and the closest signal is Accepted harmful deal x judged negotiator outgroup with a negative association ($p = 0.291$). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Empathy: Perspective taking with a positive association ($p = 0.005$). Non-parametric conclusion: no converged second-phase non-parametric model is available, so the robustness check is inconclusive for this hypothesis.

- H3. Original hypothesis: The empathy effect may vary according to whether the judged negotiator is ingroup, outgroup, or control, while decision outcome and the additional relational controls remain explicitly modeled. Tobit conclusion: the evidence is mixed but offers partial support for the hypothesis. Model A does not support the hypothesis; none of the composite empathy x judged-negotiator relational-status interactions are statistically significant, and the closest signal is Empathy x judged negotiator outgroup with a negative association ($p = 0.262$). Model B supports the hypothesis through Personal distress x judged negotiator outgroup with a positive association ($p = 0.002$) and Empathic concern x judged negotiator outgroup with a negative association ($p = 0.025$). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Empathy: Perspective taking with a positive association ($p = 0.007$). Non-parametric conclusion: no converged second-phase non-parametric model is available, so the robustness check is inconclusive for this hypothesis.

9 Hypothesis Validation and Results

Detailed coefficient tables for each Tobit model, coupled with non-parametric robustness outputs, natural language interpretive narratives, and estimator-specific diagnostics, are provided below. In the default pipeline, converged non-parametric fits immediately attempt participant-level cluster-bootstrap inference; the report labels deferred and sparse-bootstrap cases explicitly when full inference is not available.

9.1 H1: Empathy Effect

This section evaluates the primary effect of empathy on moral judgments while holding judged-negotiator status, counterpart status, and observer-side victim alignment constant.

9.1.1 Model A: Composite Empathy

Table 5: H1 Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.463	4.389	0.002**
Empathy composite (average)	-0.692	1.006	0.492
Judged negotiator outgroup (ref = ingroup)	-0.369	0.400	0.357
Judged negotiator control label hidden (ref = ingroup)	0.461	0.371	0.214
Counterpart negotiator outgroup (ref = ingroup)	-0.278	0.373	0.456
Counterpart negotiator control label hidden (ref = ingroup)	-0.103	0.356	0.773
Observer-side victim outgroup (ref = ingroup victim)	0.022	0.508	0.965
Observer role (ref = victim)	0.902	0.437	0.039*
Engineering participant (ref = humanities)	1.547	0.879	0.078+
Man (ref = woman)	0.339	0.895	0.704
Age	0.413	0.194	0.034*
Socioeconomic status	0.261	0.391	0.504
Negotiator 2 (ref = negotiator 1)	0.620	0.263	0.018*
Scale variance (log)	1.893	0.060	<0.001***

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.463 (p=0.002). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.692. The p-value of 0.492 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.369. The p-value of 0.357 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.461. The p-value of 0.214 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.278. The p-value of 0.456 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.103. The p-value of 0.773 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.022. The p-value of 0.965 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.902. The p-value of 0.039 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.547. The p-value of 0.078 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.339. The p-value of 0.704 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.413. The p-value of 0.034 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.261. The p-value of 0.504 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.893, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.953, p = 2.244e-24). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 6: H1 Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-16.504	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	-0.355	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator outgroup (ref = ingroup)	-0.146	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator control label hidden (ref = ingroup)	0.342	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator outgroup (ref = ingroup)	-0.410	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator control label hidden (ref = ingroup)	-0.007	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer-side victim outgroup (ref = ingroup victim)	0.329	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.882	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.196	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.178	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.626	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.067	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.264	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -16.504 (p=NA). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.355. The p-value of NA the p-value cannot be determined. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.146. The p-value of NA the p-value cannot be determined. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.342. The p-value of NA the p-value cannot be determined. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.410. The p-value of NA the p-value cannot be determined. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.007. The p-value of NA the p-value cannot be determined. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.329. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.882. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.196. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.178. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.626. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.067. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.1.2 Model B: Separated Empathy Constructs

Table 7: H1 Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.061	4.595	0.004**
Empathy: Fantasy scale	-0.921	0.808	0.254
Empathy: Empathic concern	-2.573	1.135	0.023*
Empathy: Perspective taking	1.722	0.858	0.045*
Empathy: Personal distress	1.312	0.966	0.175
Judged negotiator outgroup (ref = ingroup)	-0.313	0.393	0.425
Judged negotiator control label hidden (ref = ingroup)	0.389	0.365	0.287
Counterpart negotiator outgroup (ref = ingroup)	-0.143	0.357	0.688
Counterpart negotiator control label hidden (ref = ingroup)	-0.147	0.357	0.680
Observer-side victim outgroup (ref = ingroup victim)	-0.171	0.497	0.731
Observer role (ref = victim)	0.982	0.433	0.023*
Engineering participant (ref = humanities)	1.511	0.897	0.092+
Man (ref = woman)	0.233	0.947	0.805
Age	0.373	0.196	0.057+
Socioeconomic status	0.122	0.393	0.755
Negotiator 2 (ref = negotiator 1)	0.528	0.257	0.040*
Scale variance (log)	1.877	0.057	<0.001***

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.061 (p=0.004). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.921. The p-value of 0.254 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -2.573. The p-value of 0.023 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.722. The p-value of 0.045 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.312. The p-value of 0.175 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.313. The p-value of 0.425 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.389. The p-value of 0.287 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.143. The p-value of 0.688 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.147. The p-value of 0.680 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.171. The p-value of 0.731 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.982. The p-value of 0.023 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.511. The p-value of 0.092 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.233. The p-value of 0.805 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.373. The p-value of 0.057 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.122. The p-value of 0.755 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.877, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.963, p = 6.382e-22). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 8: H1 Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-16.145	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Fantasy scale	-0.678	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Empathic concern	-1.033	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Perspective taking	0.460	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Personal distress	0.612	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator outgroup (ref = ingroup)	-0.153	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator control label hidden (ref = ingroup)	0.243	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator outgroup (ref = ingroup)	-0.333	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator control label hidden (ref = ingroup)	0.062	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer-side victim outgroup (ref = ingroup victim)	0.225	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	1.061	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	0.091	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.225	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.620	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.046	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.217	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -16.145 (p=NA). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.678. The p-value of NA the p-value cannot be determined. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.033. The p-value of NA the p-value cannot be determined. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.460. The p-value of NA the p-value cannot be determined. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.612. The p-value of NA the p-value cannot be determined. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.153. The p-value of NA the p-value cannot be determined. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.243. The p-value of NA the p-value cannot be determined. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.333. The p-value of NA the p-value cannot be determined. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.062. The p-value of NA the p-value cannot be determined. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.225. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.061. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.091. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.225. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.620. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.046. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.2 H2a: Judged-Status x Decision Contrasts Without Control

This section tests whether moral judgment differs when the judged negotiator is ingroup versus outgroup and whether that contrast changes across Accept versus Reject decisions after excluding any scenario with a control-labeled negotiator.

9.2.1 Model A: Composite Empathy Control

Table 9: H2a Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	5.196	2.758	0.060+
Judged negotiator outgroup (ref = ingroup)	-0.205	0.543	0.706
Empathy composite (average)	0.095	0.664	0.887
Negotiator accepted harmful deal	-14.038	1.054	<0.001***
Counterpart negotiator outgroup (ref = ingroup)	0.027	0.379	0.944
Observer-side victim outgroup (ref = ingroup victim)	0.621	0.546	0.255
Observer role (ref = victim)	-0.110	0.533	0.836
Engineering participant (ref = humanities)	0.395	0.586	0.501
Man (ref = woman)	-0.047	0.573	0.934
Age	0.121	0.120	0.314
Socioeconomic status	0.537	0.254	0.035*
Negotiator 2 (ref = negotiator 1)	0.292	0.332	0.379
Accepted harmful deal x judged negotiator outgroup	0.050	0.681	0.942
Scale variance (log)	1.954	0.055	<0.001***

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 5.196 (p=0.060). The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.205. The p-value of 0.706 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 0.095. The p-value of 0.887 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.038. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.027. The p-value of 0.944 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.621. The p-value of 0.255 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of -0.110. The p-value of 0.836 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.395. The p-value of 0.501 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.047. The p-value of 0.934 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.121. The p-value of 0.314 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.537. The p-value of 0.035 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.050. The p-value of 0.942 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.954, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.936$, $p = 2.814\text{e-}26$). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 10: H2a Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-1.319	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator outgroup (ref = ingroup)	-0.163	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	1.473	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator accepted harmful deal	-14.849	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator outgroup (ref = ingroup)	-0.180	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer-side victim outgroup (ref = ingroup victim)	0.332	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	-0.000	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-1.098	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.308	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.406	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.519	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.251	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator outgroup	0.307	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -1.319 (p=NA). The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.163. The p-value of NA the p-value cannot be determined. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 1.473. The p-value of NA the p-value cannot be determined. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.849. The p-value of NA the p-value cannot be determined. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.180. The p-value of NA the p-value cannot be determined. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.332. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of -0.000. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -1.098. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.308. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.406. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.519. The p-value of NA the p-value cannot be determined. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.307. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.2.2 Model B: Separated Construct Controls

Table 11: H2a Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	5.333	3.041	0.079+
Judged negotiator outgroup (ref = ingroup)	-0.039	0.544	0.943
Empathy: Fantasy scale	-0.718	0.567	0.205
Empathy: Empathic concern	-1.703	0.885	0.054+
Empathy: Perspective taking	1.476	0.589	0.012*
Empathy: Personal distress	1.178	0.742	0.113
Negotiator accepted harmful deal	-13.871	1.042	<0.001***
Counterpart negotiator outgroup (ref = ingroup)	0.116	0.376	0.758
Observer-side victim outgroup (ref = ingroup victim)	0.526	0.542	0.332
Observer role (ref = victim)	-0.115	0.527	0.827
Engineering participant (ref = humanities)	0.273	0.575	0.635
Man (ref = woman)	-0.130	0.588	0.825
Age	0.092	0.127	0.468
Socioeconomic status	0.461	0.251	0.066+
Negotiator 2 (ref = negotiator 1)	0.286	0.331	0.388
Accepted harmful deal x judged negotiator outgroup	-0.086	0.660	0.896
Scale variance (log)	1.945	0.054	<0.001***

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 5.333 (p=0.079). The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.039. The p-value of 0.943 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.718. The p-value of 0.205 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.703. The p-value of 0.054 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.476. The p-value of 0.012 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.178. The p-value of 0.113 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.871. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.116. The p-value of 0.758 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.526. The p-value of 0.332 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of -0.115. The p-value of 0.827 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.273. The p-value of 0.635 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.130. The p-value of 0.825 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.092. The p-value of 0.468 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.461. The p-value of 0.066 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.086. The p-value of 0.896 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.945, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.942$, $p = 3.904e-25$). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 12: H2a Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-3.616	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator outgroup (ref = ingroup)	-0.336	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Fantasy scale	-0.647	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Empathic concern	1.143	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Perspective taking	1.373	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Personal distress	-0.083	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator accepted harmful deal	-15.533	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator outgroup (ref = ingroup)	-0.193	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer-side victim outgroup (ref = ingroup victim)	-0.186	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.244	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.925	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.462	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.461	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.608	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.359	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator outgroup	0.636	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -3.616 (p=NA). The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.336. The p-value of NA the p-value cannot be determined. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.647. The p-value of NA the p-value cannot be determined. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of 1.143. The p-value of NA the p-value cannot be determined. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.373. The p-value of NA the p-value cannot be determined. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.083. The p-value of NA the p-value cannot be determined. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -15.533. The p-value of NA the p-value cannot be determined. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.193. The p-value of NA the p-value cannot be determined. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.186. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.244. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.925. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.462. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.461. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.608. The p-value of NA the p-value cannot be determined. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.636. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.3 H2b: Judged-Status x Decision Contrasts With Control

This section examines judged-negotiator ingroup, outgroup, and control status directly, while retaining decision outcome, judged-status x decision interactions, counterpart status, and observer-side victim alignment in the full judgment sample.

9.3.1 Model A: Composite Empathy Control

Table 13: H2b Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	4.053	2.227	0.069+
Judged negotiator outgroup (ref = ingroup)	0.171	0.437	0.696
Judged negotiator control label hidden (ref = ingroup)	0.234	0.474	0.621
Empathy composite (average)	0.159	0.550	0.773
Negotiator accepted harmful deal	-14.033	0.927	<0.001***
Counterpart negotiator outgroup (ref = ingroup)	-0.146	0.284	0.608
Counterpart negotiator control label hidden (ref = ingroup)	0.176	0.285	0.536
Observer-side victim outgroup (ref = ingroup victim)	0.001	0.373	0.998
Observer role (ref = victim)	0.343	0.335	0.305
Engineering participant (ref = humanities)	1.236	0.496	0.013*
Man (ref = woman)	0.109	0.500	0.827
Age	0.149	0.100	0.137
Socioeconomic status	0.445	0.231	0.054+
Negotiator 2 (ref = negotiator 1)	0.360	0.239	0.132
Accepted harmful deal x judged negotiator outgroup	-0.506	0.597	0.396
Accepted harmful deal x judged negotiator control label hidden	0.210	0.601	0.727
Scale variance (log)	1.947	0.048	<0.001***

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 4.053 (p=0.069). The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.171. The p-value of 0.696 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.234. The p-value of 0.621 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 0.159. The p-value of 0.773 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.033. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.146. The p-value of 0.608 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.176. The p-value of 0.536 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.001. The p-value of 0.998 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.343. The p-value of 0.305 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.236. The p-value of 0.013 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.109. The p-value of 0.827 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.149. The p-value of 0.137 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.445. The p-value of 0.054 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.506. The p-value of 0.396 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.210. The p-value of 0.727 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.947, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.940, p = 1.526e-37). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 14: H2b Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-3.744	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator outgroup (ref = ingroup)	0.763	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator control label hidden (ref = ingroup)	0.375	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	0.726	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator accepted harmful deal	-14.309	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator outgroup (ref = ingroup)	-0.315	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator control label hidden (ref = ingroup)	-0.203	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer-side victim outgroup (ref = ingroup victim)	0.180	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.192	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.325	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.043	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.585	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.328	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.416	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator outgroup	-0.935	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator control label hidden	-0.177	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -3.744 (p=NA). The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.763. The p-value of NA the p-value cannot be determined. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.375. The p-value of NA the p-value cannot be determined. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 0.726. The p-value of NA the p-value cannot be determined. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.309. The p-value of NA the p-value cannot be determined. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.315. The p-value of NA the p-value cannot be determined. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.203. The p-value of NA the p-value cannot be determined. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.180. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.192. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.325. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.043.

The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.585. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.328. The p-value of NA the p-value cannot be determined. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.935. The p-value of NA the p-value cannot be determined. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.177. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.3.2 Model B: Separated Construct Controls

Table 15: H2b Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	3.915	2.400	0.103
Judged negotiator outgroup (ref = ingroup)	0.312	0.428	0.467
Judged negotiator control label hidden (ref = ingroup)	0.242	0.469	0.606
Empathy: Fantasy scale	-0.631	0.491	0.199
Empathy: Empathic concern	-1.541	0.744	0.038*
Empathy: Perspective taking	1.368	0.482	0.005**
Empathy: Personal distress	1.145	0.664	0.085+
Negotiator accepted harmful deal	-13.913	0.916	<0.001***
Counterpart negotiator outgroup (ref = ingroup)	-0.058	0.278	0.834
Counterpart negotiator control label hidden (ref = ingroup)	0.154	0.283	0.587
Observer-side victim outgroup (ref = ingroup victim)	-0.119	0.371	0.748
Observer role (ref = victim)	0.387	0.335	0.247
Engineering participant (ref = humanities)	1.137	0.499	0.023*
Man (ref = woman)	0.103	0.527	0.845
Age	0.130	0.102	0.205
Socioeconomic status	0.357	0.224	0.111
Negotiator 2 (ref = negotiator 1)	0.355	0.239	0.137
Accepted harmful deal x judged negotiator outgroup	-0.608	0.576	0.291
Accepted harmful deal x judged negotiator control label hidden	0.156	0.587	0.791
Scale variance (log)	1.939	0.048	<0.001***

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 3.915 (p=0.103). The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.312. The p-value of 0.467 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.242. The p-value of 0.606 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.631. The p-value of 0.199 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.541. The p-value of 0.038 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.368. The p-value of 0.005 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.145. The p-value of 0.085 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.913. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.058. The p-value of 0.834 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.154. The p-value of 0.587 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.119. The p-value of 0.748 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.387. The p-value of 0.247 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.137. The p-value of 0.023 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.103. The p-value of 0.845 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.130. The p-value of 0.205 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.357. The p-value of 0.111 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.608. The p-value of 0.291 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.156. The p-value of 0.791 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.939, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.942, p = 7.986e-37). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 16: H2b Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-3.643	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator outgroup (ref = ingroup)	0.760	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator control label hidden (ref = ingroup)	0.377	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Fantasy scale	-0.243	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Empathic concern	-0.513	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Perspective taking	1.060	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Personal distress	0.336	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator accepted harmful deal	-14.627	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator outgroup (ref = ingroup)	-0.331	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator control label hidden (ref = ingroup)	-0.086	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer-side victim outgroup (ref = ingroup victim)	0.125	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.415	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.079	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	-0.104	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.586	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.243	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.314	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator outgroup	-0.821	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator control label hidden	-0.125	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -3.643 (p=NA). The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.760. The p-value of NA the p-value cannot be determined. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.377. The p-value of NA the p-value cannot be determined. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.243. The p-value of NA the p-value cannot be determined. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.513. The p-value of NA the p-value cannot be determined. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.060. The p-value of NA the p-value cannot be determined. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.336. The p-value of NA the p-value cannot be determined. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.627. The p-value of NA the p-value cannot be determined. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.331. The p-value of NA the p-value cannot be determined. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.086. The p-value of NA the p-value cannot be determined. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.125. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.415. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.079. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.104. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.586. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.243. The p-value of NA the p-value cannot be determined. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.821. The p-value of NA the p-value cannot be determined. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.125. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.4 H3: Empathy x Judged-Status Moderation

This section tests whether empathy slopes differ across judged-negotiator status categories after retaining decision outcome, judged-status x decision terms, counterpart status, and observer-side victim alignment.

9.4.1 Model A: Composite Empathy Interaction

Table 17: H3 Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	3.877	2.535	0.126
Empathy composite (average)	0.235	0.617	0.703
Judged negotiator outgroup (ref = ingroup)	1.571	1.318	0.233
Judged negotiator control label hidden (ref = ingroup)	-0.511	1.510	0.735
Negotiator accepted harmful deal	-14.036	0.930	<0.001***
Counterpart negotiator outgroup (ref = ingroup)	-0.156	0.285	0.583
Counterpart negotiator control label hidden (ref = ingroup)	0.161	0.283	0.570
Observer-side victim outgroup (ref = ingroup victim)	-0.001	0.373	0.998
Observer role (ref = victim)	0.351	0.336	0.296
Engineering participant (ref = humanities)	1.233	0.497	0.013*
Man (ref = woman)	0.104	0.501	0.836
Age	0.150	0.100	0.135
Socioeconomic status	0.446	0.231	0.053+
Negotiator 2 (ref = negotiator 1)	0.347	0.237	0.142
Accepted harmful deal x judged negotiator outgroup	-0.486	0.598	0.417
Accepted harmful deal x judged negotiator control label hidden	0.203	0.601	0.735
Empathy x judged negotiator outgroup	-0.634	0.565	0.262
Empathy x judged negotiator control label hidden	0.337	0.637	0.597
Scale variance (log)	1.946	0.048	<0.001***

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 3.877 (p=0.126). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 0.235. The p-value of 0.703 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 1.571. The p-value of 0.233 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.511. The p-value of 0.735 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.036. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.156. The p-value of 0.583 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.161. The p-value of 0.570 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.001. The p-value of 0.998 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.351. The p-value of 0.296 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.233. The p-value of 0.013 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.104. The p-value of 0.836 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.150. The p-value of 0.135 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.446. The p-value of 0.053 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.486. The p-value of 0.417 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.203. The p-value of 0.735 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:judged_outgroup (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.634. The p-value of 0.262 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:judged_control (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.337. The p-value of 0.597 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.946, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.940$, $p = 1.675e\text{-}37$). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 18: H3 Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-4.030	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	1.033	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator outgroup (ref = ingroup)	2.167	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator control label hidden (ref = ingroup)	1.100	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator accepted harmful deal	-14.490	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator outgroup (ref = ingroup)	-0.343	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator control label hidden (ref = ingroup)	-0.176	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer-side victim outgroup (ref = ingroup victim)	0.279	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.176	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.303	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.010	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.574	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.297	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.427	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator outgroup	-0.664	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator control label hidden	0.049	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy x judged negotiator outgroup	-0.710	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy x judged negotiator control label hidden	-0.412	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -4.030 (p=NA). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 1.033. The p-value of NA the p-value cannot be determined. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 2.167. The p-value of NA the p-value cannot be determined. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 1.100. The p-value of NA the p-value cannot be determined. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.490. The p-value of NA the p-value cannot be determined. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.343. The p-value of NA the p-value cannot be determined. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.176. The p-value of NA the p-value cannot be determined. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.279. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.176. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.303. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.010. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.574. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.297. The p-value of NA the p-value cannot be determined. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.664. The p-value of NA the p-value cannot be determined. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.049. The p-value of NA the p-value cannot be determined. The term iri_total:judged_outgroup (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.710. The p-value of NA the p-value cannot be determined. The term iri_total:judged_control (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.412. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.4.2 Model B: Separated Constructs Interaction

Table 19: H3 Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	3.020	2.761	0.274
Empathy: Fantasy scale	-0.834	0.575	0.147
Empathy: Empathic concern	-0.857	0.830	0.302
Empathy: Perspective taking	1.792	0.667	0.007**
Empathy: Personal distress	0.438	0.694	0.528
Judged negotiator outgroup (ref = ingroup)	2.673	1.445	0.064+
Judged negotiator control label hidden (ref = ingroup)	0.127	1.804	0.944
Negotiator accepted harmful deal	-13.961	0.921	$<$0.001***
Counterpart negotiator outgroup (ref = ingroup)	-0.039	0.276	0.887
Counterpart negotiator control label hidden (ref = ingroup)	0.134	0.283	0.636
Observer-side victim outgroup (ref = ingroup victim)	-0.117	0.373	0.753
Observer role (ref = victim)	0.378	0.337	0.261
Engineering participant (ref = humanities)	1.146	0.499	0.022*
Man (ref = woman)	0.067	0.527	0.899
Age	0.139	0.103	0.180
Socioeconomic status	0.354	0.224	0.113
Negotiator 2 (ref = negotiator 1)	0.345	0.235	0.142
Accepted harmful deal x judged negotiator outgroup	-0.610	0.576	0.290
Accepted harmful deal x judged negotiator control label hidden	0.213	0.592	0.719
Fantasy x judged negotiator outgroup	-0.247	0.529	0.641
Empathic concern x judged negotiator outgroup	-1.658	0.740	0.025*
Perspective taking x judged negotiator outgroup	-0.901	0.621	0.147
Personal distress x judged negotiator outgroup	2.047	0.662	0.002**
Fantasy x judged negotiator control label hidden	0.699	0.714	0.328
Empathic concern x judged negotiator control label hidden	-0.342	0.808	0.672
Perspective taking x judged negotiator control label hidden	-0.290	0.775	0.708
Personal distress x judged negotiator control label hidden	0.107	0.664	0.872
Scale variance (log)	1.935	0.047	$<$0.001***

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 3.020 (p=0.274). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.834. The p-value of 0.147 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.857. The p-value of 0.302 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.792. The p-value of 0.007 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.438. The p-value of 0.528 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 2.673. The p-value of 0.064 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.127. The p-value of 0.944 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.961. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.039. The p-value of 0.887 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.134. The p-value of 0.636 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.117. The p-value of 0.753 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.378. The p-value of 0.261 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.146. The p-value of 0.022 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.067. The p-value of 0.899 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.139. The p-value of 0.180 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.354. The p-value of 0.113 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.610. The p-value of 0.290 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.213. The p-value of 0.719 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:judged_outgroup (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.247. The p-value of 0.641 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:judged_outgroup (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -1.658. The p-value of 0.025 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt:judged_outgroup (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.901. The p-value of 0.147 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:judged_outgroup (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 2.047. The p-value of 0.002 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:judged_control (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.699. The p-value of 0.328 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:judged_control (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.342. The p-value of 0.672 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:judged_control (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.290. The p-value of 0.708 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:judged_control (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.107. The p-value of 0.872 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.935, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.944$, $p = 1.721e\text{-}36$). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 20: H3 Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-5.464	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Fantasy scale	-0.745	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Empathic concern	0.547	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Perspective taking	1.660	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy: Personal distress	-0.128	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator outgroup (ref = ingroup)	3.399	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Judged negotiator control label hidden (ref = ingroup)	2.682	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator accepted harmful deal	-14.808	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator outgroup (ref = ingroup)	-0.277	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Counterpart negotiator control label hidden (ref = ingroup)	-0.055	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer-side victim outgroup (ref = ingroup victim)	0.057	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.267	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.137	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	-0.004	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.584	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	0.275	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.254	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator outgroup	-0.600	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Accepted harmful deal x judged negotiator control label hidden	0.192	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Fantasy x judged negotiator outgroup	0.320	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathic concern x judged negotiator outgroup	-1.888	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Perspective taking x judged negotiator outgroup	-0.657	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Personal distress x judged negotiator outgroup	1.192	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Fantasy x judged negotiator control label hidden	0.715	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathic concern x judged negotiator control label hidden	-1.349	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Perspective taking x judged negotiator control label hidden	-0.645	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Personal distress x judged negotiator control label hidden	0.367	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -5.464 (p=NA). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.745. The p-value of NA the p-value cannot be determined. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of 0.547. The p-value of NA the p-value cannot be determined. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.660. The p-value of NA the p-value cannot be determined. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.128. The p-value of NA the p-value cannot be determined. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 3.399. The p-value of NA the p-value cannot be determined. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 2.682. The p-value of NA the p-value cannot be determined. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.808. The p-value of NA the p-value cannot be determined. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.277. The p-value of NA the p-value cannot be determined. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.055. The p-value of NA the p-value cannot be determined. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of 0.057. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.267. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.137. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.004. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.584. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.275. The p-value of NA the p-value cannot be determined. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.600. The p-value of NA the p-value cannot be determined. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.192. The p-value of NA the p-value cannot be determined. The term iri_fs:judged_outgroup (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.320. The p-value of NA the p-value cannot be determined. The term iri_ec:judged_outgroup (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -1.888. The p-value of NA the p-value cannot be determined. The term iri_pt:judged_outgroup (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.657. The p-value of NA the p-value cannot be determined. The term iri_pd:judged_outgroup (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 1.192. The p-value of NA the p-value cannot be determined. The term iri_fs:judged_control (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.715. The p-value of NA the p-value cannot be determined. The term iri_ec:judged_control (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -1.349. The p-value of NA the p-value cannot be determined. The term iri_pt:judged_control (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.645. The p-value of NA the p-value cannot be determined. The term iri_pd:judged_control (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.367. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

10 Discussion and Limitations

While the dual-estimator strategy strengthens the analysis, several limitations remain. First, both inference strategies assume independence between participants, which may still be violated by broader shared institutional or contextual shocks. Second, we assume a normal distribution for the latent errors ϵ_{ij} in the Tobit branch; departures from normality could affect the consistency of the maximum likelihood estimators. The non-parametric robustness branch reduces dependence on that assumption and adds participant-level cluster bootstrap inference after converged full-sample fits, but bootstrap summaries still carry Monte Carlo error and can be sensitive to convergence problems in difficult specifications. Finally, the use of unstandardized averages for empathy assumes a linear mapping between the psychometric scale and the latent moral preference across both estimators.

11 Conclusion

Based on the combined interval-censored Tobit estimations and the cluster-aware non-parametric robustness workflow, empathy and relational judgment structure have been documented together under Option 2 through negotiator-level relational predictors rather than descriptive case labels.